

A proof of perturbative expansion for SDEs

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Abstract

We study the Langevin equation perturbed by a small potential εV , and derive a second-order asymptotic expansion of the expected value $\langle P_t^\varepsilon k \rangle$ with respect to ε . The proof is self-contained and based on SDE techniques: a variation-of-parameters (Duhamel) formula, differentiable dependence of SDE solutions on initial conditions, and $H^2(\mu)$ regularity estimates for the perturbed semigroup. In particular, the approach avoids more advanced techniques in semigroup theory or hypocoercivity, relying instead on elementary Sobolev estimates that follow directly from the Lipschitz structure of the drift. The first-order term in the expansion recovers the fluctuation-dissipation relation, while the second-order term yields an explicit expression for the leading correction to the linear response.

1 Setup

Consider, for some $\varepsilon > 0$, a stochastic process $\{X_t^\varepsilon\}_{t \geq 0} = \{(x_t^\varepsilon, v_t^\varepsilon)\}_{t \geq 0}$ following the Langevin dynamics:

$$\begin{cases} dx_t^\varepsilon = v_t^\varepsilon dt \\ dv_t^\varepsilon = -\nabla_x(U(x_t^\varepsilon) + \varepsilon V(x_t^\varepsilon))dt - v_t^\varepsilon dt + \sqrt{2}dB_t. \\ X_0^\varepsilon \sim \mu \end{cases}, \quad (1)$$

where U is a potential on $\mathbb{R}^d$ and V is (the shape of) a perturbing potential whose strength is parametrized by ε , $\{B_t\}_{t \geq 0}$ is a d -dimensional brownian motion and μ is the equilibrium probability measure associated to the unperturbed potential U ,

$$\frac{d\mu}{d\lambda}(x, v) = \frac{\exp[-U(x) - v^2/2]}{Z}.$$

We denote by $\{P_t^\varepsilon\}_{t \geq 0}$ the semigroup associated to (1) defined for $k \in L^2(\mu)$, and by $\{P_t\}_{t \geq 0}$ the semigroup corresponding to the unperturbed dynamics, i.e. (1) with $\varepsilon = 0$. $H^2(\mu)$ denotes the weighted Sobolev space corresponding to μ and averages with respect to μ are denoted by $\langle \cdot \rangle$. Set $Bk = -\nabla_x V \cdot \nabla_v k$. Our main result is:

Proposition 1.1. *For $k \in H^2(\mu)$ it holds:*

$$P_t^\varepsilon k = P_t k + \varepsilon \int_0^t P_{t-s} B P_s k ds + \varepsilon^2 \int_0^t \int_0^s P_{t-s} B P_{s-r} B P_r k dr ds + \varepsilon^3 R^\varepsilon(t),$$

with

$$\begin{aligned} R^\varepsilon(t) &= \int_0^t \int_0^s P_{t-s} B P_{s-r} S^\varepsilon(r) dr ds, \\ S^\varepsilon(t) &= \int_0^t B P_{t-s} B P_s^\varepsilon k ds \end{aligned}$$

satisfying

$$\langle R^\varepsilon(t) \rangle := \int R^\varepsilon(t) d\mu = O(1) \text{ as } \varepsilon \rightarrow 0.$$

This proposition allows us to identify the coefficients in the expansion as the first and second derivatives of the expected value $\mathbb{E}[k(X_t^\varepsilon)] = \langle P_t^\varepsilon k \rangle$:

$$\left. \frac{d}{d\varepsilon} \langle P_t^\varepsilon k \rangle \right|_{\varepsilon=0} = \int_0^t \langle B P_s k \rangle ds \quad (2)$$

$$\left. \frac{d^2}{d\varepsilon^2} \langle P_t^\varepsilon k \rangle \right|_{\varepsilon=0} = 2 \int_0^t \int_0^s \langle B P_{s-r} B P_r k \rangle dr ds. \quad (3)$$

Note that the statement is limited to expectation values, as the estimate is on the average of the remainder term $R^\varepsilon(t)$, and not on the remainder term itself. We will discuss this point in more detail in Remark 3.4.

Such an asymptotic expansion of $\langle P_t^\varepsilon k \rangle$ is of interest, because the terms on the right-hand-side involve only the dynamics corresponding to $\varepsilon = 0$ and the perturbation term. The first order term corresponds to the celebrated fluctuation-dissipation relation [8]. It might be that the unperturbed dynamics could be well known or easier to compute, or one might be able to experimentally measure it. Having an expression for the perturbed dynamics in terms of the unperturbed one thus allows gaining insight into perturbations of simple models, or allows to predict transient responses to perturbations of systems that one can measure in unperturbed (equilibrium) state. Classical examples of this principle include the Einstein relation [3] for the mobility of a particle in a fluid, or relations between magnetic susceptibilities and magnetization fluctuations in magnetic materials [7]. We note that much work has been done in recent times related to extensions of such relations to non-equilibrium situations, for instance in [1]. In terms of applications, non-equilibrium methods are particularly important to biological systems, [4, 2]. While we do not address such a general situation, given the natural formulation of non-equilibrium problems in terms of SDEs, the techniques we present here might be of use in the more general setting.

In this paper, we show a rigorous proof of an asymptotic perturbative expansion for semigroups to second order. The mature subject of semigroup theory has many rich results applying to the situation we study. Our particular goal can be achieved in the general setting where the unperturbed semigroup is quasi-bounded (i.e. $\|P_t\| \leq M e^{\omega t}$ for some $M, \omega > 0$) and the perturbation is bounded relative to the unperturbed generator [6, Chapter 9, §3.7 Asymptotic perturbation of semigroups], which in our case would read $\|\nabla_x V \cdot \nabla_v f\| \lesssim \|L f\| + \|f\|$, where L denotes the generator of the unperturbed dynamics. In this hypocoercive case, this latter condition can be checked, at least for bounded V , using results such as those in [9, Theorem 18, eq. (4.1); Section I.7]. Given the relative

sophistication of these general results, it can be of interest to have a more direct proof, based mostly on SDE techniques, which might be more familiar to a wider audience interested in problems of stochastics. In particular, we use considerably weaker regularity-preservation results (compared to what is obtained, for example, using Hörmander's approach [5]), that can be derived from the differentiable dependence of SDEs on initial conditions. This yields the restrained but relevant result stated above.

We will be working under the following

Assumption 1.2 (Assumptions on U, V).

- (i) $U \in C^3(\mathbb{R}^d)$ with bounded second and third derivatives.
- (ii) U grows at least linearly, i.e. $U(x) \geq C|x|$ for some $C > 0$.
- (iii) $V \in C^1(\mathbb{R}^d)$, bounded and with bounded first derivatives. (In particular, $L^2(\mu^\varepsilon) = L^2(\mu)$)

A few remarks on the notation: We denote by $H^n(\mu)$ the weighted Sobolev space corresponding to μ , i.e.

$$H^n(\mu) = \left\{ f : \mathbb{R}^{2d} \rightarrow \mathbb{R} \text{ measurable} : \sum_{k=0}^n \int |D^k f|^2 d\mu < \infty \right\}.$$

The semigroup $\{P_t^\varepsilon\}_{t \geq 0}$ introduced above is defined for $k \in L^2(\mu)$ as

$$(P_t^\varepsilon k)(x) = \mathbb{E}[k(X_t^{\varepsilon, x})],$$

where the process $\{X_t^{\varepsilon, x}\}_{t \geq 0}$ has the same dynamics as (1), but satisfies the initial condition $X_0^{\varepsilon, x} = x \in \mathbb{R}^{2d}$.

We denote the generator of the semigroup by L^ε and customarily drop the 0 corresponding to the unperturbed case, i.e. we write $P_t = P_t^0, L = L^0$.

We recall that

$$L^\varepsilon f = -\nabla_x(U + \varepsilon V) \cdot \nabla_x f - v \cdot \nabla_v f + \Delta f,$$

such that

$$\varepsilon Bf := (L^\varepsilon - L)f = \varepsilon(-\nabla_x V) \cdot \nabla_x f. \quad (4)$$

The remainder of this work has two more sections, one providing preliminary results mainly concerning the differentiable dependence of SDE solutions on initial conditions. These will allow us to produce, in the last section, the necessary regularity results along with the proof of the main result.

2 Preliminary results

Throughout this section, we will be considering more general SDEs for an $\mathbb{R}^n$ -valued process $\{X_t\}_{t \geq 0}$, of the form

$$dX_t = b(X_t)dt + \sigma dB_t, \quad (5)$$

with b in C^1 and Lipschitz continuous, and σ a constant matrix. In this case, it is well-known that a semigroup may be defined on continuous functions vanishing at ∞ , $C_0(\mathbb{R}^n)$. The existence of an invariant measure μ is not a-priori clear, and so we will not be talking about a semigroup defined on $L^2(\mu)$ functions. We denote the semigroup defined on $C_0(\mathbb{R}^n)$ by $\{T_t\}_{t \geq 0}$ and its generator by A .

Lemma 2.1. *Consider the solution $\{X_t^x\}_{t \geq 0}$ to a SDE as (5). Then it holds that*

$$\mathbb{E}[|X_t^x - x|^2] = (|x|^2 + 1)O(t), \quad (6)$$

and therefore also

$$\mathbb{E}[|X_t^x - x|] = (|x| + 1)O(\sqrt{t}). \quad (7)$$

By an application of Chebyshev's inequality, it then holds:

$$\mathbb{P}[|X_t^x - x| > M] < \frac{1}{M^2}(|x|^2 + 1)O(t). \quad (8)$$

All the asymptotic estimates hold as $t \rightarrow 0$.

Proof.

$$\begin{aligned} \mathbb{E}[(X_t^x - x)^2] &= \mathbb{E} \left[\left(\int_0^t b(X_s^x) ds + \int_0^t \sigma(X_s) dW_s \right)^2 \right] \\ &\leq 2\mathbb{E} \left[\left(\int_0^t b(X_s^x) ds \right)^2 \right] + 2\mathbb{E} \left[\left(\int_0^t \sigma dW_s \right)^2 \right] \\ &\leq 2\mathbb{E} \left[t \int_0^t (b(x) + b(X_s^x) - b(x))^2 ds \right] + 2C_\sigma t \\ &\leq 4t^2|b(x)|^2 + 4L_b t \int_0^t \mathbb{E}[(X_s^x - x)^2] ds + 2C_\sigma t \\ &\leq (4|b(x)|^2 + 2C_\sigma)t + 4L_b \int_0^t \mathbb{E}[(X_s^x - x)^2] ds, \text{ if } t \leq 1 \end{aligned}$$

The first inequality is because $(a + b)^2 \leq 2a^2 + 2b^2$. Next, we use Cauchy-Schwarz (for the functions 1 and $b(X_s^x)$) in the first term, and the Ito isometry in the second. In the last step, we use the same basic inequality as in the first (only in vector form) and the Lipschitz continuity of b . We can apply Gronwall's inequality, giving us

$$\mathbb{E}[(X_t^x - x)^2] \leq (4|b(x)|^2 + 2C_\sigma)te^{4L_b t},$$

which shows eq. (6). $\square$

We will use the following characterization of the domain $D(A)$ in order to argue that P_t^ε maps C_c^∞ into $D(L^\delta)$ for any $\delta \geq 0$ in Lemma 3.2.

Proposition 2.2. *For $f \in C^2(\mathbb{R}^n)$, let $\mathcal{A}$ be the differential operator corresponding to A ,*

$$\mathcal{A}f = b \cdot \nabla f + \frac{1}{2} \sigma \sigma^T : D^2 f, \quad (9)$$

where

$$\sigma \sigma^T : D^2 f = \sum_{i,j=1}^d (\sigma \sigma^T)_{ij} \partial_i \partial_j f.$$

Let D_c denote the set of functions f such that

- $f \in C_0(\mathbb{R}^n) \cap C^2(\mathbb{R}^n)$

- ∇f is bounded
- $\mathcal{A}f \in C_0(\mathbb{R}^n)$.

Then D_c is a subset of $D(A)$, and for $f \in D_c$,

$$Af = \mathcal{A}f.$$

Proof. Let $f \in D_c$. An application of Ito's formula gives:

$$f(X_t^x) - f(x) = \int_0^t b(X_s^x) \cdot \nabla f \, ds + \int_0^t \nabla f(X_s^x) \sigma \, dW_s + \frac{1}{2} \int_0^t \sigma \sigma^T(X_s^x) : D^2 f(X_s^x) \, ds.$$

Because the derivatives of f are bounded, the second term is a martingale and therefore vanishes when taking expectations. We see then that

$$\begin{aligned} \frac{1}{t} |P_t^b f(x) - f(x) - t\mathcal{A}f(x)| &= \frac{1}{t} |\mathbb{E}[f(X_t^x) - f(x)] - t\mathcal{A}f(x)| \\ &\leq \frac{1}{t} \left[\int_0^t \mathbb{E} |\mathcal{A}f(X_s^x) - \mathcal{A}f(x)| \, ds \right]. \end{aligned}$$

Denote in the following $\mathcal{A}f = g$, and recall that it is in $C_0(\mathbb{R}^{2d})$ by assumption, hence bounded and uniformly continuous. We can split the domain of the inner integration into events where $|X_s^x - x| \geq \varepsilon > 0$ and where $|X_s^x - x| \leq \varepsilon$, where ε is such that $\sup_{|a-b| < \varepsilon} |g(a) - g(b)| \leq \delta$, for some $\delta > 0$, which can be picked arbitrarily small by uniform continuity. So

$$\begin{aligned} \frac{1}{t} \left[\int_0^t \mathbb{E} |g(X_s^x) - g(x)| \, ds \right] &\leq \frac{1}{t} \int_0^t 2 \|g\|_\infty \mathbb{P}[|X_s^x - x| \geq \varepsilon] \, ds + \delta \\ &\leq \frac{1}{t} \int_0^t 2 \|g\|_\infty \mathbb{P}[|X_s^x - x| \geq \varepsilon] \, ds + \delta \\ &\leq \|g\|_\infty C(|b(x)|^2 + 1) \frac{t}{\varepsilon^2} + \delta, \end{aligned}$$

where the last step is by (8). So we can make a choice of small δ , which gives a choice of ε , and then we can choose t smaller than a threshold, such that the right-hand side is bounded by the arbitrarily small error we require.

This estimate is now uniform on compact subsets because b is continuous. In other words, we have now that for any given compact set K it holds that for any $\gamma > 0$ there exists $t_{\gamma,K} > 0$ such that

$$\frac{1}{t} \left[\int_0^t \mathbb{E} |g(X_s^x) - g(x)| \, ds \right] < \gamma,$$

for all $t \leq t_{\gamma,K}, x \in K$.

We now need to provide an estimate for when x is outside of a certain compact set. We make use of the fact that $g \in C_0(\mathbb{R}^{2d})$, and so it is small for large x .

Let now $R > 1$ such that for $|x| > R$, $|g(x)| < \gamma/4$. If we start the process at x with $|x| > 2R$, $|g(X_s^x) - g(x)| > \gamma/2$ only if $|X_s^x - x| > |x|/2$, as otherwise $|X_s^x| > |x| - |X_s^x - x| > |x|/2 > R$. So we have

$$\begin{aligned} \frac{1}{t} \left[\int_0^t \mathbb{E} |g(X_s^x) - g(x)| ds \right] &\leq 2 \|g\|_\infty \frac{1}{t} \int_0^t \mathbb{P} [|X_s^x - x| > |x|/2] ds + \gamma/2 \\ &\leq \frac{1}{t} 2 \|g\|_\infty C \frac{(|b(x)|^2 + 1)}{|x|^2/4} \int_0^t s ds + \gamma/2 \\ &\leq C' \|g\|_\infty \frac{|x|^2 + 1}{|x|^2} t + \gamma/2 \leq 2C' \|g\|_\infty t + \gamma/2, \end{aligned}$$

where in the second to last step we used that $|b(x)| \leq |b(0)| + L_b|x|$. So, given $\gamma > 0$, first choose R as above, set $K = \bar{B}_{2R}$ and then for $t < \min \left\{ t_{\gamma,K}, \frac{\gamma}{2C'\|g\|_\infty} \right\}$ it holds

$$\frac{1}{t} \left[\int_0^t \mathbb{E} |g(X_s^x) - g(x)| ds \right] < \gamma,$$

for all $x \in \mathbb{R}^d$. □

Proposition 2.3. *(Differentiable dependence on initial conditions) Let $x \in \mathbb{R}^n$ be a starting condition, $z \in \mathbb{R}^n$ be a unit vector, $t \geq 0$ and $h > 0$, and set*

$$\alpha_t^{x,z,h} = \frac{X_t^{x+zh} - X_t^x}{h}.$$

Then $\lim_{h \rightarrow 0} \alpha_t^{x,z,h}$ exists in $L^2(\mathbb{P})$ and is equal to the random variable $\alpha_t^{x,z}$ given by

$$\begin{cases} d\alpha_t^{x,z} = Db(X_t^x)[\alpha_t^{x,z}] dt \\ \alpha_0^{x,z} = z. \end{cases} \quad (10)$$

Furthermore, almost surely, $\alpha^{x,z,h}$, and hence also $\alpha_t^{x,i}$ are bounded as:

$$\alpha^{x,z,h} \leq e^{L_b t}; \alpha_t^{x,z} \leq e^{L_b t}, \text{ for all } x \in \mathbb{R}^d, t \geq 0, \quad (11)$$

where L_b is the Lipschitz constant of b .

Also, $\alpha_t^{x,z}$ has a continous dependence on x , that is

$$\lim_{y \rightarrow x} \alpha_t^{y,z} = \alpha_t^{x,z}, \quad (12)$$

almost surely and, by the boundedness of $\alpha_t^{x,i}$, also in $L^2(\mathbb{P})$.

Proof. We begin by noting that $\alpha_t^{x,z}$ is well defined for almost all fixed ω , as the solution to the ODE eq. (10), where the driving field $F(t, y) = Db(X_t^x)[y]$ is Lipschitz in the “space” coordinate y , with constant L_b .

Turning to the approximation $\alpha^{x,z,h}$, it holds, by replacing X_t^x and X_t^{x+zh} from the SDE they satisfy:

$$\begin{aligned} \alpha^{x,z,h} &= z + \int_0^t \frac{1}{h} [b(X_s^{x+zh}) - b(X_s^x)] ds, \text{ and therefore} \quad (\diamond) \\ |\alpha^{x,z,h}| &\leq 1 + L_b \int_0^t |\alpha^{x,z,h}| ds, \end{aligned}$$

from which an application of Gronwall's inequality gives

$$|\alpha^{x,z,h}| \leq e^{L_b t}. \quad (\star)$$

From eq. (◇) and by replacing $\alpha_t^{x,z}$ from its equation,

$$\alpha^{x,z,h} - \alpha_t^{x,z} = \int_0^t \left(\frac{1}{h} [b(X_s^{x+zh}) - b(X_s^x)] - Db(X_s^x)[\alpha_s^{x,z}] \right) ds. \quad (\circ)$$

By Taylor's theorem,

$$\begin{aligned} b(X_s^{x+zh}) - b(X_s^x) &= Db(X_s^x)[X_s^x - X_s^x] + R(X_s^x, X_s^{x+zh}), \text{ with} \\ R(X_s^x, X_s^{x+zh}) &= o(h) \text{ as } h \rightarrow 0, \end{aligned}$$

with the constant in the asymptotic bound $o(h)$ possibly depending on ω , but being uniform in $s \in [0, t]$.

The reason for this is that the error term $R(X_s^x, X_s^{x+zh})$ is bounded by a sum of errors of the form

$$(D(b_i)(\zeta_i(X_s^x, X_s^{x+zh})) - D(b_i)(X_s^x))[X_s^{x+zh} - X_s^x], \quad i = 1 \dots d,$$

with $\zeta_i(X_s^x, X_s^{x+zh})$ being a convex combination of X_s^x, X_s^{x+zh} as given by the mean value theorem. Because of eq. (★), $\zeta_i(X_s^x, X_s^{x+zh}) \rightarrow X_s^x$ uniformly in s and hence, by the uniform continuity of Db on compact domains,

$$D(b_i)(\zeta_i(X_s^x, X_s^{x+zh})) - D(b_i)(X_s^x) = o(1),$$

and using eq. (★) again, we have the error $o(h)$ in the Taylor approximation.

Plugging this into eq. (◇), we have

$$\alpha^{x,z,h} - \alpha_t^{x,z} = \int_0^t (Db(X_s)[\alpha_s^{x,z,h} - \alpha_s^{x,z}] + o(1)) ds,$$

from which Gronwall's inequality gives

$$|\alpha^{x,z,h} - \alpha_t^{x,z}| \leq o(1)e^{L_b t},$$

showing that for almost all fixed ω , and all fixed x, t ,

$$\alpha_t^{x,z,h} \rightarrow \alpha_t^{x,z} \text{ as } h \rightarrow 0.$$

The bound (★) implies eq. (11) and the convergence in $L^2(\mu)$.

To show the continuous dependence of $\alpha_t^{x,z}$ on x , we consider $y \in \mathbb{R}^n$ and write

$$\alpha_t^{y,z} - \alpha_t^{x,z} = \int_0^t (Db(X_s^y)[\alpha_s^{y,z}] - Db(X_s^x)[\alpha_s^{x,z}]) ds.$$

We can subtract and add $Db(X_s^y)[\alpha_s^{x,z}]$ inside the integral, giving

$$\begin{aligned} |\alpha_t^{y,z} - \alpha_t^{x,z}| &\leq \int_0^t |(Db(X_s^y) - Db(X_s^x))[\alpha_s^{x,z}]| ds + \int_0^t |Db(X_s^y)[\alpha_s^{y,z} - \alpha_s^{x,z}]| ds \\ &\leq 2tL_b e^{L_b t} + L_b \int_0^t |\alpha_s^{y,z} - \alpha_s^{x,z}| ds, \end{aligned}$$

where we used eq. (11) and the boundedness of Db in the last step. Equation (12) then follows by Gronwall's inequality. $\square$

We will from now on write $\alpha_t^{x,i}$ instead of α_t^{x,e_i} , where e_i is the i -th canonical basis vector in $\mathbb{R}^n$. The following lemma will be used in particular for functions in $C_c^\infty(\mathbb{R}^n)$:

Lemma 2.4. *Let $f \in C^1(\mathbb{R}^n)$ be bounded and have bounded, uniformly continuous derivatives. Then $T_t f \in C^1(\mathbb{R}^n)$ and*

$$\partial_i T_t f(x) = \mathbb{E} \left[Df(X_t^x) [\alpha_t^{x,i}] \right], \quad (13)$$

with $\alpha_t^{x,i}$ as in Proposition 2.3.

If, additionally, f has bounded and uniformly continuous second derivatives and Db is also Lipschitz-continuous, then $T_t f \in C^2(\mathbb{R}^n)$ and

$$\partial_j \partial_i T_t f(x) = \mathbb{E} \left[D^2 f(X_t^x) [\alpha_t^{x,i}, \alpha_t^{x,j}] + Df(X_t^x) [\beta_t^{x,i,j}] \right], \quad (14)$$

where $\beta_t^{x,i,j} \in \mathbb{R}^n$ satisfies the equation:

$$\begin{cases} d\beta_t^{x,i,j} = D^2 b(X_t^x) [\alpha_t^{x,i}, \beta_t^{x,i,j}] dt \\ \beta_0^{x,i,j} = e_j \end{cases} \quad (15)$$

Remark 2.5. By letting

$$F(x, y) = Df(x)[y]$$

and $\{\mathcal{T}_t\}_{t \geq 0}$ be the semigroup associated to the joint SDE for the X and α variables:

$$\begin{cases} dX_t^x = b(X_t^x) dt + \sigma dB_t \\ d\alpha_t^{x,z} = Db(X_t^x) [\alpha_t^{x,z}] dt \\ X_0^x = x \\ \alpha_0^{x,z} = z \end{cases}, \quad (16)$$

we could write the expression (13) as

$$\partial_i T_t f(x) = \mathbb{E} \left[Df(X_t^x) [\alpha_t^{x,i}] \right] = (\mathcal{T}_t F)(x, e_i).$$

However, F is not a bounded function, so it is not clear that $\mathcal{T}_t F$ makes sense. We can easily address this issue by noting that, due to eq. (11), F can be replaced by

$$F_t(p, q) = Df(p) [\chi_t(|q|)q],$$

where $\chi_t \in C_c^\infty(\mathbb{R})$ with $\chi_t(\rho) = 1$ for $\rho \leq e^{L_b t}$.

Proof. (Of Lemma 2.4) We have that, for a unit vector $z \in \mathbb{R}^d$, $h > 0$

$$\begin{aligned} \frac{1}{h} [T_t f(x + zh) - T_t f(x)] &= \mathbb{E} \left[\frac{f(X_t^{x+zh}) - f(X_t^x)}{h} \right] \\ &= \mathbb{E} \left[\int_0^1 Df(\theta X_t^x + (1-\theta)X_t^{x+zh}) \left[\frac{X_t^{x+zh} - X_t^x}{h} \right] d\theta \right] \\ &= \int_0^1 \mathbb{E} \left[Df(\theta X_t^x + (1-\theta)X_t^{x+zh}) \left[\frac{X_t^{x+zh} - X_t^x}{h} \right] \right] d\theta, \end{aligned}$$

where integrals may be exchanged due to the boundedness of integrands, by assumption for Df and by eq. (11).

Using the convergence in probability of X_t^{x+zh} to X_t^x (a consequence of Proposition 2.3) along with the boundedness and uniform continuity of Df , it follows that $Df(\theta X_t^x + (1-\theta)X_t^{x+zh}) \rightarrow Df(X_t^x)$ in $L^2(\mathbb{P})$ as $h \rightarrow 0$. Along with the convergence of $\frac{X_t^{x+zh} - X_t^x}{h} \rightarrow \alpha_t^{x,z}$ given by Proposition 2.3, this implies that the integrand of the outer integral converges to $\mathbb{E}[Df(X_t^x)[\alpha_t^{x,z}]]$ for all θ and, as it is bounded uniformly in h , it follows that

$$\lim_{h \rightarrow 0} \frac{T_t f(x + zh) - T_t f(x)}{h} = \mathbb{E}[Df(X_t^x)[\alpha_t^{x,z}]], \quad (17)$$

pointwise in x . The continuity of the function follows from the continuous dependence on the initial condition of both X_t^x and $\alpha_t^{x,z}$ given by Proposition 2.3.

For the second part, use Remark 2.5 to write

$$\partial_i T_t \phi(x) = (\mathcal{T}_t F_t)(x, e_i).$$

We can apply Proposition 2.3 to eq. (16) to see that

$$\lim_{h \rightarrow 0} \frac{\left(X_t^{x+he_j}, \alpha_t^{x+he_j, i} \right) - \left(X_t^x, \alpha_t^{x, i} \right)}{h} = (\alpha_t^{x, j}, \beta_t^{x, i, j}).$$

It is easily seen that

$$DF(x, \alpha)[(\delta_x, \delta_\alpha)] = D^2 f(x)[\alpha, \delta_x] + Df(x)[\delta_\alpha],$$

and we can apply the proof of the first part, now to $\mathcal{T}_t$ and F_t , to obtain

$$\begin{aligned} \partial_j \mathcal{T}_t F_t &= \mathbb{E}[DF_t(X^x, \alpha_t^{x, i})[\alpha_t^{x, j}, \beta_t^{x, i, j}]] = \mathbb{E}[DF(X^x, \alpha_t^{x, i})[\alpha_t^{x, j}, \beta_t^{x, i, j}]] \\ &= \mathbb{E} \left[D^2 f(X_t^x)[\alpha_t^{x, i}, \alpha_t^{x, j}] + Df(X_t^x)[\beta_t^{x, i, j}] \right], \end{aligned}$$

where we were able to replace F_t by F , due to the boundedness of $\alpha_t^{x, i}$. $\square$

3 Regularity estimates and main result

Lemma 3.1.

(i) Let $k \in H^1(\mu)$. Then $P_t^\varepsilon k \in H^1(\mu)$ and

$$\|\partial_i P_t^\varepsilon k\|_{L^2(\mu)} \leq e^{C_\varepsilon t} \|\nabla k\|_{L^2(\mu)}, \quad (18)$$

where C_ε is the Lipschitz constant of the drift term $(-\nabla_x(U + \varepsilon V), -v)$, which is in particular uniformly bounded for ε in some interval around 0.

(ii) If furthermore $k \in H^2(\mu)$, $P_t^\varepsilon k \in H^2(\mu)$ and

$$\|\partial_j \partial_i P_t^\varepsilon k\|_{L^2(\mu)} \leq C_2(\varepsilon, t) \left(\|\nabla k\|_{L^2(\mu)} + \|D^2 k\|_{L^2(\mu)} \right), \quad (19)$$

where C_2 is again bounded on any set of the form $[0, \delta] \times [0, T]$.

Proof. Throughout the proof, we drop the ε for ease of notation in P_t^ε and $X_t^{x,\varepsilon}$, so that they become P_t and X_t^x , respectively.

Let $\phi \in C_c^\infty(\mathbb{R}^n)$. Equations (11) and (13) give:

$$\begin{aligned} |\partial_i P_t \phi| &= |\mathbb{E}[Df(X_t^x)[\alpha_t^{x,i}]]| \\ &\leq e^{C_\varepsilon t} \mathbb{E}[|\nabla f|(X_t^x)], \end{aligned}$$

so that

$$\|\partial_i P_t \phi\|_{L^2(\mu)} \leq e^{C_\varepsilon t} \|\nabla f\|_{L^2(\mu)}. \quad (20)$$

Therefore, P_t is a continuous map on the dense subset C_c^∞ of the complete space $H^1(\mu)$. Thus, it extends to the whole space, with the bound (20) being preserved, and the extension equals P_t , since the latter is already well-defined on the larger space $L^2(\mu)$.

For item 2, one starts from eq. (14) and, using the boundedness of $\alpha^{x,i}, \alpha^{x,j}, \beta^{x,i,j}$, which depends continuously on ε and t , one proceeds analogously to item 1. $\square$

Lemma 3.2. *Let $\varepsilon, \delta \geq 0$ and let $D(L^\delta)$ be the domain of the unbounded operator L^δ . For $k \in C_c^\infty$ it holds $P_t^\varepsilon k \in D(L^\delta)$.*

Proof. We show that $P_t^\varepsilon k \in D(A^\delta) \subset D(L^\delta)$. In order to show this, we want to apply Proposition 2.2. Lemma 2.4 gives that $P_t^\varepsilon k \in C^2(\mathbb{R}^{2d})$ with bounded derivatives.

We additionally need to show that $\mathcal{A}^\delta P_t^\varepsilon k \in C_0(\mathbb{R}^{2d})$. Since $T_t^\varepsilon k \in D(A^\varepsilon) \cap C^2(\mathbb{R}^{2d})$, the consideration of the more general characteristic operator, as discussed in section 7.5 of [10], allows us to conclude that $\mathcal{A}^\varepsilon T_t^\varepsilon k = A^\varepsilon T_t^\varepsilon k$. Therefore, $\mathcal{A}^\varepsilon P_t^\varepsilon k = \mathcal{A}^\varepsilon T_t^\varepsilon k = A^\varepsilon T_t^\varepsilon k = T_t^\varepsilon A^\varepsilon k \in C_0(\mathbb{R}^{2d})$.

Next, $\mathcal{A}^\delta = \mathcal{A}^\varepsilon - (\delta - \varepsilon) \nabla_x V \cdot \nabla_v$, so what we need to check is that $\partial_{v_i} P_t^\varepsilon k \in C_0(\mathbb{R}^{2d})$. The continuity of the function follows from Lemma 2.4. From (13) and (11) we see that

$$|\partial_i P_t^\varepsilon k(x)| = \left| \mathbb{E}[Dk(X_t^{x,\varepsilon})[\alpha_t^{x,i,\varepsilon}]] \right| \leq e^{Ct} \mathbb{E}[|\nabla k|(X_t^{x,\varepsilon})] = e^{Ct} P_t^\varepsilon |\nabla k|(x),$$

establishing that $\partial_{v_i} P_t^\varepsilon k$ vanishes at infinity, since P_t^ε maps $|\nabla k| \in C_c^\infty(\mathbb{R}^{2d}) \subset C_0(\mathbb{R}^{2d})$ into $C_0(\mathbb{R}^{2d})$. $\square$

The basic step used for arriving at the asymptotic expansion is the following ‘‘Variation of parameters’’ formula:

Lemma 3.3. *(Variation of parameters) For $k \in H^2(\mu)$:*

$$P_t^\varepsilon k = P_t k + \varepsilon \int_0^t P_{t-s} B P_s^\varepsilon k ds, \quad (21)$$

$$\nabla_v P_t^\varepsilon k = \nabla_v P_t k + \varepsilon \int_0^t \nabla_v P_{t-s} B P_s^\varepsilon k ds. \quad (22)$$

Furthermore, the function

$$S^\varepsilon(r) = \int_0^r B P_{r-s} B P_s^\varepsilon k ds$$

is in $H^1(\mathbb{R}^{2d})$ for all $t \geq 0$, and $\|S(t)\|_{L^2(\mu)}$ is $O(1)$ as $\varepsilon \rightarrow 0$, locally uniformly in time.

Proof. We will show eq. (21) first $k \in C_c^\infty$. By Lemma 3.1 (i), the mapping $k \mapsto \int_0^t P_{t-s} B P_s^\varepsilon k$ is continuous on $H^1(\mu)$, so the result for C_c^∞ extends to $H^1(\mu) \supset H^2(\mu)$, by density. For $k \in C_c^\infty$, we can write $P_t^\varepsilon k - P_t k = P_{t-s} P_s^\varepsilon k|_{s=0}^t$. For $\phi \in L^2(\mu)$ one then has

$$\langle \phi(P_t^\varepsilon k - P_t k) \rangle = \langle \phi(P_{t-s} P_s^\varepsilon k) \rangle|_{s=0}^t.$$

Using the chain rule and Lemma 3.2, one checks that

$$\frac{d}{ds} \langle \phi P_{t-s} P_s^\varepsilon k \rangle = \langle \phi P_{t-s} (L - L^\varepsilon) P_s^\varepsilon k \rangle,$$

and therefore

$$\langle \phi(P_t^\varepsilon k - P_t k) \rangle = \int_0^t \langle \phi P_{t-s} (L^\varepsilon - L) P_s^\varepsilon k \rangle ds,$$

Next we use Lemma 3.1 (i) which together with the boundedness of $\nabla_x V$ implies that the function $s \mapsto P_{t-s} (L^\varepsilon - L) P_s^\varepsilon k$ is in $L^2([0, T], L^2(\mu))$, allowing us to identify $\int_0^t P_{t-s} (L^\varepsilon - L) P_s^\varepsilon k ds$ on the r.h.s., that is:

$$\langle \phi(P_t^\varepsilon k - P_t k) \rangle = \left\langle \phi \int_0^t P_{t-s} (L^\varepsilon - L) P_s^\varepsilon k ds \right\rangle,$$

establishing eq. (21).

In order to check eq. (22), we need to show that we can bring the ∇_v inside the integral. For $\phi \in C_c^\infty$, and letting $\partial_{v_i}^*$ be the formal adjoint of ∂_{v_i} obtained by partial integration, we have

$$\begin{aligned} \langle \phi \partial_{v_i} (P_t^\varepsilon - P_t) k \rangle &= \langle \partial_{v_i}^* \phi (P_t^\varepsilon - P_t) \rangle = \langle \partial_{v_i}^* \phi \int_0^t P_{t-s} B P_s^\varepsilon k ds \rangle \\ &= \int_0^t \langle \partial_{v_i}^* \phi P_{t-s} B P_s^\varepsilon k \rangle ds \\ &= \int_0^t \langle \phi \partial_{v_i} P_{t-s} B P_s^\varepsilon k \rangle ds. \end{aligned} \quad (\diamond)$$

If we consider the function $f_i = s \mapsto \partial_{v_i} P_{t-s} B P_s^\varepsilon k$, we can use Lemma 3.1 (ii) to see that $B P_s^\varepsilon k$ is in $H^1 \mu$ with a uniform bound on the $H^1(\mu)$ norm, and then use Lemma 3.1 (i) to see that

$$\|f_i(s)\|_{L^2(\mu)} \leq C(\varepsilon, t) \|f\|_{H^2 \mu}, \quad (\star)$$

for a constant $C(\varepsilon, t)$ bounded on any set of the form $[0, \delta] \times [0, T]$.

Therefore, $\int_0^t f_i(s) ds$ is well-defined as an $L^2(\mu)$ function, and we can write eq. (◇) as

$$\int_0^t \langle \phi \partial_{v_i} P_{t-s} B P_s^\varepsilon k \rangle ds = \left\langle \phi \int_0^t f_i(s) ds \right\rangle,$$

establishing eq. (22).

Multiplying (22) by $(-\nabla_x V) \cdot$ and using that $\nabla_x V$ is bounded, we get

$$\frac{1}{\varepsilon} (-\nabla_x V) \cdot [\nabla_v P_t^\varepsilon k - \nabla_v P_t k] = \int_0^t B P_{t-s} B P_s^\varepsilon k ds = \sum_{i=1}^d \int_0^t (-\partial_{x_i} V) f_i ds =: S^\varepsilon(t). \quad (23)$$

The l.h.s. is, by Lemma 3.1 (ii), a $H^1(\mu)$ function. The desired bound on S^ε follows from (★). $\square$

In the following remark, we highlight a rather fortituous situation that allows the proof of the proposition above, because we believe that this kind of argument might be of use in other settings.

Remark 3.4. *We have shown in Lemma 3.3 that $S^\varepsilon(r)$ and (by Lemma 3.1) $P_{s-r}S^\varepsilon(r)$ are in $H^1(\mu)$. The $L^2(\mu)$ part of the norm corresponding to the derivatives of $S^\varepsilon(t)$ are however not easily shown to satisfy an estimate other than $O(\varepsilon^{-1})$, which is not sufficient for our purposes, as $S^\varepsilon(r)$ is multiplied by $O(\varepsilon^3)$ in the expansion of $P_t^\varepsilon k$ (see eqs. (24) and (25)).*

It was however at hand to show $\|S^\varepsilon(r)\|_{L^2(\mu)} = O(1)$ (Lemma 3.3). We then intend to get around the fact that a derivative is being applied to $P_{s-r}S^\varepsilon(t)$ in

$$\langle R^\varepsilon(t) \rangle = \int_0^t \int_0^s \langle P_{t-s}(-\nabla_x V) \cdot \nabla_v P_{s-r}S^\varepsilon(r) \rangle dr ds,$$

by an integration by parts.

We can drop the first P_{t-s} , because the average is relative to the stationary measure μ . The integration by parts then gives

$$\begin{aligned} \langle (-\nabla_x V) \cdot \nabla_v P_{s-r}S^\varepsilon(r) \rangle &= \int_{\mathbb{R}^{2d}} (-\nabla_x V) \cdot \nabla_v P_{s-r}S^\varepsilon(r) \rho(x, v) dx dv \\ &= \langle \nabla_x V \cdot (\nabla_v \ln \rho) P_{s-r}S^\varepsilon(r) \rangle. \end{aligned}$$

The uniform estimate now follows, because the functions $\partial_{v_i} \ln \rho = -v_i$ are in $L^2(\mu)$, and because of the $L^2(\mu)$ -estimate on $S^\varepsilon(r)$.

In summary, the proof of Proposition 1.1 crucially depends on the stationarity of the measure, allowing to drop the P_{t-s} , and on the fact that $\partial_{v_i} \ln \rho \in L^2(\mu)$, allowing to get around the absence of an $H^1(\mu)$ -estimate through an integration by parts and usage of an $L^2(\mu)$ estimate.

Proof. (Of Proposition 1.1) We replace $\nabla_v P_s^\varepsilon k$ in eq. (21) using eq. (22):

$$\begin{aligned} P_t^\varepsilon k &= P_t k + \varepsilon \int_0^t P_{t-s}(-\nabla_x V) \cdot \left(\nabla_v P_s k + \varepsilon \int_0^s \nabla_v P_{s-r} B P_r^\varepsilon k \right) dr ds \\ &= P_t k + \varepsilon \int_0^t P_{t-s}(-\nabla_x V) \cdot \nabla_v P_s k ds \\ &\quad + \varepsilon^2 \int_0^t P_{t-s}(-\nabla_x V) \cdot \int_0^s \nabla_v P_{s-r} B P_r^\varepsilon k dr ds, \end{aligned} \tag{24}$$

where we can use linearity to split the outer integral because by Lemma 3.1 (i) $\nabla_v P_s k$ is uniformly bounded in $L^2(\mu)$, so that the integral of $s \mapsto P_{t-s}(-\nabla_x V) \cdot \nabla_v P_s k$ is defined.

Next, using eq. (22), we see that

$$\begin{aligned} \nabla_v P_{s-r} B P_r^\varepsilon k &= \nabla_v P_{s-r}(-\nabla_x V) \cdot \nabla_v P_r^\varepsilon k \\ &= \nabla_v P_{s-r}(-\nabla_x V) \cdot \left(\nabla_v P_r k + \varepsilon \int_0^r \nabla_v P_{r-q} B P_q^\varepsilon k dr \right). \\ &= \nabla_v P_{s-r} B P_r k + \varepsilon \nabla_v P_{s-r} S^\varepsilon(r). \end{aligned} \tag{25}$$

Replacing in the integral over $[0, s]$ in eq. (24), we have

$$\begin{aligned}\int_0^s \nabla_v P_{s-r} B P_r^\varepsilon k dr &= \int_0^s [\nabla_v P_{s-r} B P_r k + \varepsilon \nabla_v P_{s-r} S^\varepsilon(r)] dr \\ &= \int_0^s \nabla_v P_{s-r} B P_r k dr + \varepsilon \int_0^s \nabla_v P_{s-r} S^\varepsilon(r) dr\end{aligned}$$

where we have used Lemma 3.1 (ii) and (i) to see that the function $r \mapsto \nabla_v P_{s-r} B P_r k$ is in $L^2([0, s], L^2(\mu))$, justifying splitting of the integral.

Replacing in the second order term of eq. (24), we get

$$\begin{aligned}&\varepsilon^2 \int_0^t P_{t-s}(-\nabla_x V) \cdot \int_0^s \nabla_v P_{s-r} B P_r^\varepsilon k dr ds \\ &= \varepsilon^2 \int_0^t P_{t-s}(-\nabla_x V) \cdot \int_0^s \nabla_v P_{s-r} B P_r k dr ds \\ &\quad + \varepsilon^3 \int_0^t P_{t-s}(-\nabla_x V) \cdot \int_0^s \nabla_v P_{s-r} S^\varepsilon(r) dr ds \\ &= \varepsilon^2 \int_0^t \int_0^s P_{t-s} B P_{s-r} B P_r k dr ds + \varepsilon^3 \int_0^t \int_0^s P_{t-s} B P_{s-r} S^\varepsilon(r) dr ds,\end{aligned}$$

where in the last step, we have moved some applications of the semigroup and the bounded operator $\nabla_x V \cdot$ inside the integrals.

The estimate then follows due to Remark 3.4. □

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